

An Alternating Coordinate-Block Counterexample to a Banach Feichtinger Conjecture

Discovery packet for arXiv:2301.03365

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Status. Full counterexample, pending human review. Conjecture 2.5 is false under the strict diagonal-dominance formulation in Definition 2.4 and also under the usual Riesz-sequence reading. The construction is not localized and leaves the paper’s Theorem 2.7 intact.

1 Source conjecture and terminology

Krishna’s *Localized Bounded Below Approximate Schauder Frames are Finite Unions of Approximate Riesz Sequences*, arXiv:2301.03365 [1], calls a pair $(\{f_n\}, \{\tau_n\})$ an approximate Schauder frame (ASF) when

$$Sx = \sum_{n=1}^{\infty} f_n(x)\tau_n$$

defines a bounded invertible operator. It is *bounded below* when $\inf_n |f_n(\tau_n)| > 0$.

Definition 2.4 asks for a finite partition $\mathbb{N} = Q_1 \cup \dots \cup Q_r$ such that every class obeys

$$\inf_{n \in Q_j} \left(|f_n(\tau_n)| - \sum_{\substack{m \in Q_j \\ m \neq n}} |f_n(\tau_m)| \right) > 0. \quad (1)$$

Conjecture 2.5 states that every bounded-below ASF admits the proposed finite decomposition. The conjecture itself prints “ARBs,” which is not defined in the paper, while the title, abstract, section heading, Definition 2.4, and Theorem 2.7 use “ARSs.” The proof of Theorem 2.7 establishes exactly (1). We therefore refute this precise displayed condition and also check the ordinary Riesz-sequence interpretation.

2 Proof intuition

An ASF only requires the *ordered* rank-one series to converge to an invertible operator. This permits large finite cancellation patterns whose supports escape to infinity. On the k th coordinate, use an odd block of $2k + 1$ copies of the same vector, with coefficient functionals carrying alternating signs. The block sum is one coordinate projection, so the full frame operator is the identity. But the block sizes are unbounded. Any finite coloring repeats a color inside some block; that same-color pair has off-diagonal interaction of modulus one, equal to the diagonal, and destroys strict diagonal dominance. The identical vectors also rule out a Riesz sequence independently of the source’s terminology.

$$\inf_{n \in \mathbb{N}} |f_n(\tau_n)| > 0.$$

Motivated from the definition of localization of frames [20][24], we introduce the following notion.

Definition 2.3. An ASF $(\{f_n\}_n, \{\tau_n\}_n)$ for $\mathcal{X}$ is said to be **intrinsically/self localized** if there exist $s > 1$ and $A > 0$ such that

$$|f_n(\tau_m)| \leq \frac{A}{(1 + |n - m|)^s}, \quad \forall n, m \in \mathbb{N}.$$

Two notions of Riesz sequences for Banach spaces exist in literature, see [115] and [32]. Here we define another.

Definition 2.4. An ASF $(\{f_n\}_n, \{\tau_n\}_n)$ for $\mathcal{X}$ is said to be an **approximate Riesz sequence** (we write ARS) if there exists a finite partition $Q_1, \dots, Q_N$ of $\mathbb{N}$ such that

$$\mathbb{N} = \bigcup_{j=1}^N Q_j$$

and for each $1 \leq j \leq N$,

$$\inf_{n \in Q_j} \left(|f_n(\tau_n)| - \sum_{m \in Q_j, m \neq n} |f_n(\tau_m)| \right) > 0.$$

Note that for Hilbert spaces, if $\{f_n\}_n$ is determined by $\{\tau_n\}_n$ (Riesz representation), then Definition 2.4 is equivalent (due to positivity) to the definition of Riesz sequence (see [23]). We now formulate the following conjecture (some other are formulated in [31]).

Conjecture 2.5. Every bounded below ASF can be partitioned as a finite union of ARBs.

We now prove Conjecture 2.5 for intrinsically localized ASFs with the help of following result.

Theorem 2.6. [23]

(i) For every $s > 1$,

$$D_s := \sup_{x \in \mathbb{R}} \sum_{n=1}^{\infty} \frac{1}{(1 + |n - x|)^s} < \infty.$$

(ii) For every $s > 1$, there exists a $C_s > 0$ (which does not depend on δ) such that

Figure 1: Definition 2.4 and Conjecture 2.5 on page 2 of the source paper.

3 The counterexample

Let $(e_k)_{k \geq 1}$ be the standard orthonormal basis of $\ell_2(\mathbb{N})$, with coordinate functionals e_k^* . Partition $\mathbb{N}$ into the consecutive odd blocks

$$I_k = \{k^2, k^2 + 1, \dots, (k+1)^2 - 1\}, \quad |I_k| = 2k + 1.$$

For $n \in I_k$, define

$$\tau_n = e_k, \quad f_n = (-1)^{n-k^2} e_k^*. \quad (2)$$

Theorem 1. The pair in (2) is a bounded-below ASF for $\ell_2(\mathbb{N})$ with frame operator the identity, but no finite partition of $\mathbb{N}$ satisfies (1). Moreover, no finite partition makes every vector subsequence $(\tau_n)_{n \in Q_j}$ a Riesz sequence.

Proof. Write P_k for the projection onto $\text{span}\{e_1, \dots, e_k\}$ and $P_0 = 0$. For $x \in \ell_2(\mathbb{N})$ and $n \in I_k$,

$$f_n(x)\tau_n = (-1)^{n-k^2} x_k e_k.$$

The signs on I_k are $+1, -1, +1, \dots, -1, +1$. Consequently their total is one, and every initial sum within that block is either zero or one. It follows that every frame partial sum ending inside I_k equals either $P_{k-1}x$ or $P_k x$. Since $P_k x \rightarrow x$ in ℓ_2 , the ordered series converges for every x and

$$Sx = \sum_{n=1}^{\infty} f_n(x)\tau_n = x.$$

Thus $S = I$ is bounded and invertible. Also

$$|f_n(\tau_n)| = 1 \quad (n \in \mathbb{N}),$$

so the ASF is bounded below with lower value one.

Now consider an arbitrary partition $\mathbb{N} = Q_1 \cup \dots \cup Q_r$. Choose k with $2k + 1 > r$. By the pigeonhole principle there exist distinct $n, m \in I_k$ in one class, say Q_j . Since $\tau_n = \tau_m = e_k$,

$$|f_n(\tau_n)| = 1, \quad |f_n(\tau_m)| = 1.$$

Therefore

$$|f_n(\tau_n)| - \sum_{\substack{q \in Q_j \\ q \neq n}} |f_n(\tau_q)| \leq 0,$$

and the infimum required in (1) cannot be positive. Since the partition was arbitrary, no finite partition works.

Finally, the same indices have identical vectors $\tau_n = \tau_m$. The synthesis map on their class sends the nonzero finitely supported coefficient vector with entries 1 at n and -1 at m to zero. Hence that vector subsequence is not a Riesz sequence (or Riesz basic sequence) under the usual injectivity/minimality requirement. No finite partition can avoid this collision in all blocks. $\square$

4 Scope and adversarial checks

The example does not satisfy intrinsic localization. Indeed, within I_k one has $|f_n(\tau_m)| = 1$ for all n, m , while the block diameters $2k$ tend to infinity. Thus no fixed $A > 0$ and $s > 1$ can give the paper’s decay bound $A/(1 + |n - m|)^s$. This is why the counterexample refutes the general Conjecture 2.5 without contradicting its localized Theorem 2.7.

The construction also exposes the exact role of conditional ordering. The partial sums are uniformly bounded coordinate projections, so the ASF definition is met without ambiguity, but alternating signs allow arbitrarily large finite redundancy. Stronger hypotheses such as unconditionality or factorization through an ℓ_p analysis space are not claimed here.

The script `verify_finite_blocks.py` checks the block sums and partial sums exactly and exhausts small coloring cases. It is a sanity check; the general proof is the argument above.

5 Novelty and review recommendation

A bounded search on 13 August 2026 covered the local run indexes and parsed arXiv corpus; exact conjecture and title phrases; the ASF/ARS terminology; and web searches for Conjecture 2.5, arXiv:2301.03365 with “counterexample/refute/false,” and bounded-below ASF counterexamples. The results found the target and mirrors but no prior counterexample or explicit resolution. This does not establish exhaustive novelty.

A human reviewer should focus on (i) whether ordered convergence is indeed the intended ASF convention in Definition 2.1; (ii) the scope chosen in light of the source’s ‘ARB’ typo; and (iii) the bounded novelty search. Human review has not yet been completed.

References

- [1] K. Mahesh Krishna, *Localized Bounded Below Approximate Schauder Frames are Finite Unions of Approximate Riesz Sequences*, arXiv:2301.03365 (2023).